EPSR-test-ex99 Miscellaneous data-set and run options

Program(s): epsr | chained passes: 3 | verification: ALL IDENTICAL (all 3 passes)

What this case tests

Exercises a group of per-data-set and global options of the `epsr` engine of EPSR [1,2] that a user commonly adjusts: Q limits, background-subtraction radius, data scaling, energy weighting of data sets, resolution, the R-space coefficient and the input Q scale [3].

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `runepsr17` (`qmin`, `rshmin`, `tweak`, `efilereq` -> `efacm`, `rspcrmin`/`rspcfrac`, `qinscalefc`)
- `epsr_standalone_rev3.f`: `inita` (`fwhm`), `ftback` (`rshmin`), `ptgnsq` (`qminp`, minimum radius of the EP)
- `epsr_standalone_rev3.f`: `get_wts`, `invert_cwt` (rows weighted by `efacm`)

Rust (port, rust/epsr-core/src/):

- `epsr.rs`: `Epsr::setup`, `Epsr::iterate`, `write_results`
- `structure.rs`: `inita`, `ftback`; `potential.rs`: `ptgnsq`
- `weights.rs`: `get_wts`, `invert_cwt`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å^3). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. For this case data set 3 gets zero weight in the empirical potential (`efilereq` 0) and the three data sets are scaled by 1.03, 1.0 and 0.97.

Inputs

Only the `.EPSR.inp` differs from EPSR-test (`scripts/make_inp.sh` via `scripts/setkw.py`): `qmin` `0.60` `0.80`; `rshmin` 0.5 / 0.9 / 1.1; `tweak` 1.03 / 1.0 / 0.97; `efilereq` 1.0 / 0.5 / 0.0; `fwhm` `0.03`; `rspcrmin` `1.50`; `rspcfrac` `0.3`; `qinscalefc` `0.98` `0.5`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 3 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 3 passes.
Evidence from the README: qminm = 0.60 raises `nqmin` from 8 to 13; the log reports minimum radius integers of the potential 19 / 33 / 39 from `rshmin`; `efacm` = `efilereq/mean(nonzero efilereq)` scales the data-set rows of the inversion matrix, which is recalculated whenever `efacm` changes; `rspcrmin`/`rspcfrac` change the R-space coefficient and the `.erg` trend line.

Notes

`qinscalefc` `0.98` `0.5` has a first value < 1, so coarse graining stays off: the values are only read and written back (`selfbroadpwr` is only used with coarse graining).

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.